AYESH UL HASSAN

Game Developer | Unreal Engine 5 | Roblox
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SUMMARY

I am a game developer working across Unreal Engine 5 and Roblox Studio, specialising in player controllers, stealth mechanics, and NPC AI systems in UE5 alongside VFX and UI/UX systems in Roblox. My shipped horror-survival game, Locked In Fear, has already attracted 500+ wish listed users before its full release - built end-to-end in Unreal Engine 5 with C++, AI Perception, collision systems, and level design, backed by a 34-page Game Design Document, proving I can take a project from concept to a live product with real-world traction. At Zehn Studio 26, I own the VFX pipeline and UI/UX design for Vermilian Mask, a live Roblox project. I bring hands-on experience in Blueprint Visual Scripting, performance profiling, Git, Flutter, and Roblox Studio/Lua, backed by internships and studio experience delivering production-ready systems under real deadlines.

WORK EXPERIENCE

Junior Roblox Developer | Zehn Studio 26 - Vermilian Mask | *May 2026 – Present*

- Sole VFX Artist on Vermilian Mask - designed and implemented every visual effect in the project end-to-end, including ability, impact, and environmental effects, owning the game's VFX pipeline from concept through in-Studio integration.
- UI/UX systems designed across multiple Roblox titles - built menus, HUDs, and interaction flows tailored to Roblox Studio's UI framework, improving usability and visual consistency across the studio's game portfolio.
- Dual VFX and UI/UX ownership - combined visual effects and interface design responsibilities within a single production pipeline, ramping up on Roblox Studio and Lua-based tooling within the first three months on the team.

Unreal Engine Intern | CUSIT Battle Zone - Investigation 51 | *June 2026 – July 2026*

- 60 FPS target achieved on GTX 1080-class hardware by refactoring tightly coupled Blueprint systems into modular Actor Components, eliminating the bottlenecks and material overflow errors that were causing frame drops and build instability.
- Blueprint-to-C++ migration reduced processing overhead and sped up iteration cycles by introducing a hybrid architecture that replaced custom Blueprint-only implementations with optimised C++ gameplay systems.
- First-person camera systems rebuilt - redesigned animation timelines and attachment setups for the player controller, delivering smoother movement and more responsive first-person interactions across all states.
- Codebase maintainability improved by restructuring tightly coupled gameplay dependencies into decoupled Actor Components, making systems significantly easier to extend as Investigation 51 continues development.

Application Development Intern | Dev-Spire Solutions | *July 2025 - August 2025*

- 6 production-ready screens shipped in 6 weeks - redesigned the full Courier Management System UI/UX in Figma (yellow-themed design system), implemented in Flutter with responsive layouts across Home, Contact, and Admin Panel dashboards, backed by SQL data management with fast, no-delay retrieval.
- Clean Git workflow maintained throughout - managed source code on GitHub in collaboration with the senior developer, tracking all sprint milestones and delivering consistent version history across the full internship window.

Gameplay Programmer - Final Year Project | City University of Science & IT | *2025 - 2026*

- 500+ wishlisted users attracted before full release - shipped Locked In Fear (lockedinfear.me), a UE5 horror-survival game, within a 6-month academic deadline alongside a 3-person team, with the public website live and promotional gameplay content available.

- GTX 1050-class performance unlocked - resolved critical frame drops and texture overflow issues on RTX 3060 hardware by converting tightly coupled Blueprint systems to C++, achieving stable frame rates on mid-range hardware after full optimisation.
- Multi-layer enemy AI built using Unreal's AI Perception Component - implemented sight detection, hearing detection, and live microphone input mechanics so the AI reacts to both player actions and environmental sounds, creating a genuine stealth-survival loop.
- All core systems owned end-to-end - delivered stealth mechanics, collision systems, level design, and player controller logic from pre-production through to live deployment on a public domain, guided by a 34-page Game Design Document.

LEADERSHIP & EXTRACURRICULAR

Advisor to President | YPDC-CUSIT Chapter & Debating Society | *December 2025 - Present*

- 3 major university events co-organised - including the 150 Years of Quaid-e-Azam Celebration, International Mother's Day Event, and Kashmir Day programs, managing logistics, student coordination, seating, and certificate distribution for each.

PROJECTS

Locked In Fear - UE5 Horror Survival | lockedinfear.me | *54-page GDD available on request*

- C++ and Blueprint gameplay systems: player controller, AI Perception (sight + hearing + microphone input), stealth mechanics, collision systems, level design. Stable on GTX 1050-class hardware after optimisation. 500+ wish listed users pre-release.

Investigation 51 - UE5 Survival Horror (In Development) | notion.so/Fading-Echoes-341205e08f9a826980f381413e03d828

- RE7-inspired first-person survival game. Actively refactoring Blueprint systems into modular C++ Actor Components, targeting 60 FPS on GTX 1080-class hardware. Full production workspace maintained in Notion - tracking assets, Blender work, material references, SketchFab sources, and to-dos.

Essence - Web Platform

- Full-stack fragrance community platform with impression mapping and a moderated marketplace, built with Firebase and Supabase data management and responsive web development.

SKILLS

Game Engines: Unreal Engine 5, Roblox

Languages: C++, Blueprint Visual Scripting, Dart / Flutter, SQL

UE5 Systems: Actor Components, AI Perception, AI Behavior Trees, Animation Blueprint, Player Controller, Collision & Physics, Performance Profiling, Level Design

Dev Tools: Git / GitHub, Figma, Notion, Blender (asset referencing)

Documentation: Game Design Documents (GDD), production wikis, asset tracking

Soft Skills: Self-directed project delivery, cross-functional teamwork, iterative playtesting, deadline management

EDUCATION

B.CS in Software Engineering | City University of Science and IT, Peshawar | Oct 2022 - Present

F.SC in Computer Science | Peshawar Model Degree College | 2020 - 2022

SELECTED ACHIEVEMENTS

- Selected for 5 competitive tech events across 3 years - Black Byt3 Convention at Islamia College (2026), Skillathon CUSIT (2024 & 2026), Cyber Fest at Peshawar University (2025), and Azure Web App Deployment Workshop by MLSA (2026).
- NAVTTC Graphic Designing Certificate + IT Diploma - Xpert IT Solution (2022-2023).